2021 2nd International Conference on Artificial Intelligence and Education (ICAIE) | 978-1-6654-2492-9/21/$31.00 ©2021 IEEE | DOI: 10.1109/ICAIE53562.2021.00143

Students' Learning Performance Evaluation Using a New Fuzzy Inference System

Shuang Wen*
School of Information Engineering
Guangdong University of Technology, GDUT
Guangzhou, China
* Corresponding author: ws2134_co@163.com

Dongfeng Liu
School of Information Engineering
Guangdong University of Technology, GDUT
Guangzhou, China
liudf@gdut.edu.cn

***Abstract*—The research of computer technology in the field of education makes Intelligent Tutoring System (ITS) birth and develop rapidly. The main advantage of an ITS is it can provide appropriate teaching content according to students' knowledge state, level and ability. Reasonable evaluation of students' learning performance is the key and difficult point in ITS design. Using fuzzy logic to evaluate students' learning performance is not a very new method. However, most approaches rely extremely on student's answer time for each question, which are seldom collected in common exams. Given this insight, we present a new fuzzy inference mechanism in this paper to calculate students' ranking order without dependence of the answer times. According to the importance, complexity and difficulty of the problem, new reasoning rules are set up to accurately evaluate the comprehensive level of problem. The results of the case study show that each student's grades are distinguished, even for those with the same grades. The proposed method in this paper enriches the present evaluation approach of students' performance, and is utilizable to analyze the direct relationship of students' scores with the attributes of each question.**

Keywords-learning performance evaluation; fuzzy rules; fuzzy inference

I. INTRODUCTION

The improvement of students' learning performance is one of the ultimate goals of developing educational systems. A high quality evaluation system can ensure that all students receive relatively fair evaluation that can truly reflect the learning situation of students, and will not hinder the students' future prospects. Thus, in order to effectively enhance students' learning performance, we first have to scientifically and accurately evaluate the students' performance, and find the factors which have a greatest influence than others. Evaluation can be used to c positively motivate students to study well, in turn, encourage teachers to teach more passionately.

Arithmetic and statistics approaches constitute the classic method to evaluating students' learning performance, such as calculating the total score of all questions. The calculated results based on this method usually include that some students have equal scores. It is difficult to say that students with the same score have the same learning level. For example, in such a common situation, if a student completes only part of a difficult problem and another student completes a simple problem completely, the score they rewarded is equal. How can we carefully distinguish the difference between the two. Therefore, we infer that fair ranking order of students' performance should not be obtained from simple statistical grades but also consider the attributes of questions, e.g. importance, complexity and difficulty [2][3][4].

Fuzzy evaluation method for applying fuzzy logic and reasoning has proven beneficial to serve the student academic performance evaluation [5][7][15], which has some advantages over classic method introduced above. In [3], Bai and Chen put forward a new fuzzy method for the shortcomings of [21], which can discriminate the students with the same grades without affecting the original ranking order. But both methods have a flaw that they rely too much on the student's response time. As we know, it's hard to get the response time for each question in most exams, and it is not necessary to get it because students may do half of one question to do another, and then come back finish it. So the methods above have obvious limits in practical applications.

In this paper, a fuzzy inference based system for learning achievement evaluation is proposed. In the process of inferring, three importance attitudes: importance, complexity, and difficulty are considered for each question. 25 fuzzy rules are designed to implement the fuzzy reasoning and further to construct seven levels for each question.

The remainder of this paper is organized as follows. We give a brief overview of fuzzy inference and it application in student's learning performance in Section 2. The proposed model and algorithm are introduced in Section 3. The numerical results for a case study are introduced in Section 4. Conclusions are drawn in Section 5.

II. LITERATURE REVIEW

Fuzzy logic was initially presented by Zadeh in 1965(e.g. [1]), and has become a widely used and tested effective technique in reasoning, so called fuzzy reasoning. All fuzzy set theory is defined on the concept of a membership function (MF), we often use fuzzy sets and membership functions indiscriminately. The fuzzy set membership functions enable the interpretations of linguistic variables express and solve various problems in a very natural and rational way. Nowadays, fuzzy set theory has reached a wide range of application in the field of education to rationalize learning evaluation of students.

The work of Biswas [5] is based on a vector approach. Each question in the answerscript is awarded a discrete fuzzy set, and then compared with a standard fuzzy set. A total score is obtained by a weighted average. His work is easy to understand

978-1-6654-2492-9/21/$31.00 ©2021 IEEE
DOI 10.1109/ICAIE53562.2021.00143

Authorized licensed use limited to: Tejasvi Kolhe. Downloaded on July 12,2026 at 18:07:37 UTC from IEEE Xplore. Restrictions apply.

and implement. Two approaches of using fuzzy sets are proposed in [8] to solve the problem of assigning two different linguistic variables to the same points. In [20], Wang and Chen introduced the concept of degrees of confidence between 0 and 1 to comprehensively evaluate students' answerscipts.

Because today's education is around the students rather than teachers, in order to improve the students' participation degree, Ma and Zhou [16] incorporates student's opinions into assessment criteria. For overcome the shortcoming that the triangle MFs employed in the original fuzzy evaluation system relies too much on the preset parameters, Gaussian MFs are introduced to improve the stability of system [13]. Yadav and Singh [18] proposed some practical approaches by using fuzzy logic technology to evaluate students' performance, and compared with the existing statistical methods.

Three attributes of questions: difficulty, importance and complexity were found to be very important in the evaluation of students' learning achievement [21]. Based on the thoughts of Weon and Kim [21], some optimization methods were presented for the fuzzy evaluation of learning performance, for the purposes: (1) sort the students with the same score reasonably [3][12]; (2) automatically constructing grade MFs [4]; (3) automatically generating the weights of the attributes of fuzzy rules, including accuracy rate, time rate, difficulty, complexity, answer-cost and importance [9].

Some works presented some practical application of fuzzy evaluation systems. Konstantina and Maria [14] have evaluated the effectiveness and accuracy of the student model in the context of online computer programming instruction. Wilson and Freeman [22] proposed an automatic scoring system using fuzzy reasoning, and combined genetic algorithms to adapt to particular instructor's attitude. Yadav. S. R, and Singh P. V. [18] studied a fuzzy expert system, which can automatically calculate test scores of semester-1 and semester-2 to get the final evaluation at the end of the academic year. Chai et al proposed a fuzzy ranking algorithm for peer assessment of students [6]. Some performance indices are obtained reflecting a student's contribution in a group. Gokmen et al in [11] studied the performance evaluation in laboratory applications. A set of lab–based examinations have been designed and a student evaluation enhancement fuzzy logic system to analyze the students' learning performance has been implemented [17].

In a short, there are two drawbacks in current works for fuzzy evaluation of students' learning performance: (1) intensively depending on the test response time to each question, which is not available for most evaluation systems; (2) the evaluation process is very complex, and the evaluation results usually can't reflect the design intention of teachers and domain experts, which in turn helps to improve the examination design.

III. METHOD

Fuzzy environment is a kind of situation related to fuzziness, imprecision and lack of information of some elements for reasoning or calculation. The fuzzy set theory can be used to define the features of observation and screening. For example, the learning performance can be expressed as linguistics: very weak, weak, medium, strong, very strong and so on. Next, we will elaborate a new fuzzy reasoning method that does not depend on the reaction time of students in the following five parts.

A. Data Prepared

Suppose there are m questions in a test paper and n students participate in the test. Let Q_i denotes the ith question and S_j denote the jth student, where $1 \le i \le m$ and $1 \le j \le n$. The accuracy rate matrix A representing the test results, is shown as following,

$$A = \begin{matrix} Q_1 \\ Q_2 \\ \vdots \\ Q_m \end{matrix} \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$$

where $a_{ij} \in [0,1]$ indicates the score of student j on question i divided by the maximum score assigned to question i. The assigned score distribution of the test questions is represented by G, $G=[g_i]$, $g_i \in [1,100]$, where:

$$\sum_{i=1}^{m} g_i = 100. \tag{1}$$

The classical ranking is to add up the scores of all the questions of each student and then sort them, is acquired using formula as follows:

$$S = G^T A = [s_j] \quad 1 \le j \le n \tag{2}$$

where $s_j \in [0,100]$ is the total score of student j.

We consider a question's three aspect: importance , complexity and difficult. Importance indicates the degree to which students should master the knowledge points involved in this problem. Complexity, relates the kind of thinking and knowledge, and how much ways needed to solve a problem. Both importance and complexity of a question can be quantitatively described with a crisp value. Thus, for a list of questions, the importance and complexity can be given as a vector of crisp values by a domain expert,

$$PV = \begin{matrix} Q_1 \\ Q_2 \\ \vdots \\ Q_m \end{matrix} \begin{bmatrix} p_1 \\ p_2 \\ \vdots \\ p_m \end{bmatrix} \qquad CV = \begin{matrix} Q_1 \\ Q_2 \\ \vdots \\ Q_m \end{matrix} \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_m \end{bmatrix}$$

where PV is importance vector, and CV is complexity vector.

Difficulty refers to how much effort it takes to solve a problem, which is determined by the accuracy rate and students' answering time [2][3][18]. However, in most examinations, the answer time for a specific question is usually not recorded. Especially for the cases in China, the paper written exams allows students to complete the questions in a whole time, for example in two hours, during which time the student can freely choose any questions to answer. Therefore, the answer time is completely unavailable to teachers. Considering this, we regard the difficulty of a question is only related to the students' accuracy rate and if a student gets a higher accuracy rate, the difficulty of the problem is lower, and vice versa. The difficulty of question i relative to student j is denoted by d_{ij}, which is defined by the corresponding accuracy rate a_{ij} of the question.

Authorized licensed use limited to: Tejasvi Kolhe. Downloaded on July 12,2026 at 18:07:37 UTC from IEEE Xplore. Restrictions apply.

$$d_{ij} = 1 - a_{ij}. \quad (3)$$

The difficulty of each question is determined by the average of all students' difficulty, so we have a difficulty vector *DV* for all questions,

$$DV = \begin{matrix} Q_1 \\ Q_2 \\ \vdots \\ Q_m \end{matrix} \begin{bmatrix} d_1 \\ d_2 \\ \vdots \\ d_m \end{bmatrix} \qquad d_i = \frac{\sum_{j=1}^{n} d_{ij}}{n}, 1 \le i \le m. \quad (4)$$

B. Evaluation Process

Once we have the data prepared as above, we can begin the process of learning performance evaluation as follows:

Step1: Establish the fuzzy domain of question

The three attributes of the question, i.e. importance, difficulty and complexity, are fuzzified by using their membership functions of fuzzy set. Triangular and trapezoidal fuzzy numbers are adopted and defined as 3-tuple (α,a,β) and 4-tuple(α,a,b,β),respectively, and the membership function of $\tilde{A}$ fuzzy set is defined as follows,

$$\mu\tilde{A}(\chi) = \begin{cases} 1 - \dfrac{a - \chi}{\alpha} & a - \alpha \le \chi \le a \\ 1 - \dfrac{\chi - a}{\beta} & a \le \chi \le a + \beta \\ 0 & othewise \end{cases} \quad (5)$$

$$\mu\tilde{A}(\chi) = \begin{cases} 1 - \dfrac{a - \chi}{\alpha} & a - \alpha \le \chi \le a \\ 1 & a \le \chi \le b \\ 1 - \dfrac{\chi - a}{\beta} & a \le \chi \le b + \beta \\ 0 & othewise \end{cases} \quad (6)$$

where α, a, b and β are real numbers and $\alpha \le a \le b \le \beta$.

Each attribute of each question is divided into three levels. Three linguistic variables and corresponding fuzzy numbers are presented in Table 1, and their membership functions are shown in Fig.1.

TABLE I. LINGUISTIC VARIABLES AND FUZZY NUMBERS FOR IMPORTANCE

Linguistic Variables	*Fuzzy Numbers*
Low	(0,0,0.2,0.4)
Medium	(0.2,0.5,0.8)
High	(0.6,0.8,1,1)

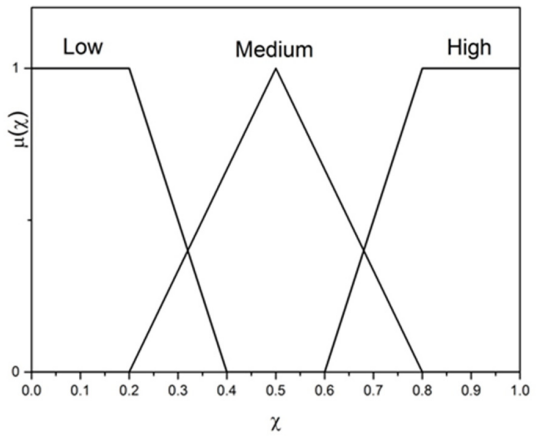


Figure 1. Membership functions of the fuzzy stets "low", "medium", and ''high", respectively

Step2: Fuzzification of questions' attribute and comprehensive level

Fuzzify the crisp values of importance, complexity and difficulty vectors. Based on the importance vector *PV*, complexity vector *CV*, difficulty vector *DV*, we calculate the fuzzy importance matrix *PM*, complexity matrix *CM*, and difficulty matrix *DM*,

$$PM = \begin{matrix} Q_1 \\ Q_2 \\ \vdots \\ Q_m \end{matrix} \begin{bmatrix} p_{11} & p_{12} & \cdots & p_{1l} \\ p_{21} & p_{22} & \cdots & p_{2l} \\ \vdots & & \ddots & \vdots \\ p_{m1} & p_{m2} & \cdots & p_{ml} \end{bmatrix}$$

$$CM = \begin{matrix} Q_1 \\ Q_2 \\ \vdots \\ Q_m \end{matrix} \begin{bmatrix} c_{11} & c_{12} & \cdots & c_{1l} \\ c_{21} & c_{22} & \cdots & c_{2l} \\ \vdots & & \ddots & \vdots \\ c_{m1} & c_{m2} & \cdots & c_{ml} \end{bmatrix}$$

$$DM = \begin{matrix} Q_1 \\ Q_2 \\ \vdots \\ Q_m \end{matrix} \begin{bmatrix} d_{11} & d_{12} & \cdots & d_{1l} \\ d_{21} & d_{22} & \cdots & d_{2l} \\ \vdots & & \ddots & \vdots \\ d_{m1} & d_{m2} & \cdots & d_{ml} \end{bmatrix}$$

where l is the number of fuzzy sets, equal to 3 in this situation, $p_{ik} \in [0,1]$ denoting the membership degree of the importance degree of the question Q_i in fuzzy set k, $c_{ik} \in [0,1]$ denoting the membership degree of the complexity of the question Q_i in fuzzy set k, $d_{ik} \in [0,1]$ denoting the membership degree of the difficulty of question Q_i in fuzzy set k.

The output variable is the comprehensive level of the question. Seven fuzzy sets are employed: Very Low, Low, More or Less Low, Medium, More or Less High, High and Very High. Their corresponding membership functions are illustrated in Figure 2.

Step3: Rules and inference generation

The rules determine the relationship between input fuzzy sets and output fuzzy sets are set up according to Table 2, it will be used in inference process, and described in details from Rule R1 to Rule R25.

Authorized licensed use limited to: Tejasvi Kolhe. Downloaded on July 12,2026 at 18:07:37 UTC from IEEE Xplore. Restrictions apply.

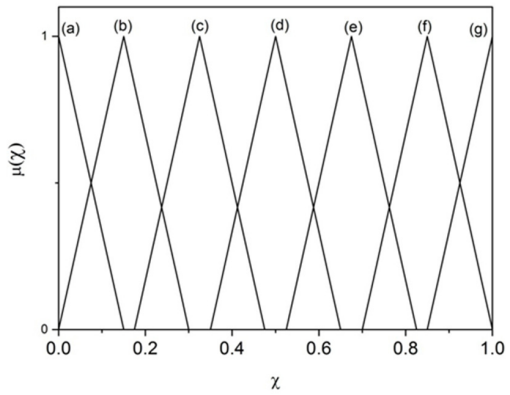


Figure 2. Membership functions of the questions' comprehensive level

TABLE II. FUZZYY RULES

Comprehensive Level	***Importance***	***Complexity***	***Difficulty***
Very Low	Low	Low	Low
Low	Medium	Low	Low
	Low	Medium	Low
	Low	Low	Medium
More or Less Medium	Medium	Medium	Low
	Medium	Low	Medium
	Low	Medium	Medium
	Low	Low	High
	Low	High	Low
	High	Low	Low
Medium	Medium	Medium	Medium
	Low	Medium	High
	Medium	High	Low
	High	Low	Medium
	Medium	Low	High
	Low	High	Medium
	High	Medium	Low
More or Less High	Medium	Medium	High
	Medium	High	Medium
	High	Medium	Medium
	High	High	Low
	High	Low	High
	Low	High	High
High	High	High	Medium
	High	Medium	High
	Medium	High	High
Very High	High	High	High

The fuzzy rules that infer the comprehensive level of question Q_i ($1 \le i \le m$) is "Very Low" is shown as follows:

Rule R1: IF importance is Low, complexity is Low, difficulty is Low THEN question's comprehensive level is Very Low.

The formula that calculating comprehensive level of question Q_i is "Very Low" is shown as follows:

$$v_{i1} = 0.4 * p_{i1} + 0.3 * c_{i1} + 0.3 * d_{i1}. \quad (7)$$

The fuzzy rules that infer the comprehensive level of question Q_i is "Low" are shown in the following:

Rule R2: IF importance is Medium, complexity is Low, difficulty is Low THEN comprehensive level is Low.

Rule R3: IF importance is Low, complexity is Medium, difficulty is Low THEN comprehensive level is Low.

Rule R4: IF importance is Low, complexity is Low, difficulty is Medium THEN comprehensive level is Low.

The formula for level "Low" is shown in the following:

$$v_{i2}=Max(0.4*p_{i2}+0.3*c_{i1}+0.3*d_{i1},0.4*p_{i1}+0.3*c_{i2}+0.3*d_{i1}, 0.4*p_{i1}+0.3*c_{i1}+0.3*d_{i2}). \quad (8)$$

The fuzzy rules that infer the comprehensive level of question Q_i is "More or Less Medium" are shown in the following:

Rule R5: IF importance is Medium, complexity is Medium, difficulty is Low THEN comprehensive level is More or Less Medium.

Rule R6: IF importance is Medium, complexity is Low, difficulty is Medium THEN comprehensive level is More or Less Medium.

Rule R7: IF importance is Low, complexity is Medium, difficulty is Medium THEN comprehensive level is More or Less Medium.

Rule R8: IF importance is Low, complexity is Low, difficulty is High THEN comprehensive level is More or Less Medium.

Rule R9: IF importance is Low, complexity is High, difficulty is Low THEN comprehensive level is More or Less Medium.

Rule R10: IF importance is High, complexity is Low, difficulty is Low THEN comprehensive level is More or Less Medium.

The formula for level "More or Less Medium" is shown in following:

$$v_{i3}=Max(0.4*p_{i2}+0.3*c_{i2}+0.3*d_{i1},0.4*p_{i2}+0.3*c_{i1}+0.3*d_{i2}, 0.4*p_{i1}+0.3*c_{i2}+0.3*d_{i2},0.4*p_{i1}+0.3*c_{i1}+0.3*d_{i3}, 0.4*p_{i1}+0.3*c_{i3}+0.3*d_{i1},0.4*p_{i3}+0.3*c_{i1}+0.3*d_{i1}). \quad (9)$$

Authorized licensed use limited to: Tejasvi Kolhe. Downloaded on July 12,2026 at 18:07:37 UTC from IEEE Xplore. Restrictions apply.

The fuzzy rules that infer the comprehensive level of question Q_i is "Medium" are shown in the following:

Rule R11: IF importance is Medium, complexity is Medium, difficulty is Medium THEN comprehensive level is Medium.

Rule R12: IF importance is Low, complexity is Medium, difficulty is High THEN comprehensive level is Medium.

Rule R13: IF importance is Medium, complexity is High, difficulty is Low THEN comprehensive level is Medium.

Rule R14: IF importance is High, complexity is Low, difficulty is Medium THEN comprehensive level is Medium.

Rule R15: IF importance is Medium, complexity is Low, difficulty is High THEN comprehensive level is Medium.

Rule R16: IF importance is Low, complexity is High, difficulty is Medium THEN comprehensive level is Medium.

Rule R17: IF importance is High, complexity is Medium, difficulty is Low THEN comprehensive level is Medium.

The formula for level "Medium" is shown in the following:

$$v_{i4}=Max(0.4*p_{i2}+0.3*c_{i2}+0.3*d_{i2},0.4*p_{i1}+0.3*c_{i2}+0.3*d_{i3},\ 0.4*p_{i2}+0.3*c_{i3}+0.3*d_{i1},0.4*p_{i3}+0.3*c_{i1}+0.3*d_{i2},\ 0.4*p_{i2}+0.3*c_{i1}+0.3*d_{i3},0.4*p_{i1}+0.3*c_{i3}+0.3*d_{i2},\ 0.4*p_{i3}+0.3*c_{i2}+0.3*d_{i1}) \quad (10)$$

The fuzzy rules that infer the comprehensive level of question Q_i is "More or Less High" are shown in the following:

Rule R18: IF importance is Medium, complexity is Medium, difficulty is High THEN comprehensive level is More or Less High.

Rule R19: IF importance is Medium, complexity is High, difficulty is Medium THEN comprehensive level is More or Less High.

Rule R20: IF importance is High, complexity is Medium, difficulty is Medium THEN comprehensive level is More or Less High.

Rule R21: IF importance is High, complexity is High, difficulty is Low THEN comprehensive level is More or Less High.

Rule R22: IF importance is High, complexity is Low, difficulty is High THEN comprehensive level is More or Less High.

Rule R23: IF importance is Low, complexity is High, difficulty is High THEN comprehensive level is More or Less High.

The formula for level "More or Less High" is shown in the following:

$$v_{i5}=Max(0.4*p_{i2}+0.3*c_{i2}+0.3*d_{i3},0.4*p_{i2}+0.3*c_{i3}+0.3*d_{i2},\ 0.4*p_{i3}+0.3*c_{i2}+0.3*d_{i2},0.4*p_{i3}+0.3*c_{i3}+0.3*d_{i1},\ 0.4*p_{i3}+0.3*c_{i1}+0.3*d_{i3},0.4*p_{i1}+0.3*c_{i3}+0.3*d_{i3}). \quad (11)$$

The fuzzy rules that infer the comprehensive level of question Q_i is "High" are shown in the following:

Rule R24: IF importance is High, complexity is High, difficulty is Medium THEN comprehensive level is High.

Rule R25: IF importance is High, complexity is Medium, difficulty is High THEN comprehensive level is High.

Rule R26: IF importance is Medium, complexity is High, difficulty is High THEN comprehensive level is High.

The formula for level "High" is shown in the following:

$$v_{i6}=Max(0.4*p_{i3}+0.3*c_{i3}+0.3*d_{i2},0.4*p_{i3}+0.3*c_{i2}+0.3*d_{i3},0.4*p_{i2}+0.3*c_{i3}+0.3*d_{i3}). \quad (12)$$

The fuzzy rule that infer the comprehensive level of question Q_i is " Very High" is shown in the following:

Rule R27: IF importance is High, complexity is High, difficulty is High THEN comprehensive level is Very High.

The formula for level "Very High" is shown in the following:

$$v_{i7}=0.4*p_{i3}+0.3*c_{i3}+0.3*d_{i3}. \quad (13)$$

By using (7) to (13) iteratively, we can get the final comprehensive levels of all questions, which is represented by the matrix LM,

$$\mathrm{LM}=\begin{matrix}\mathrm{Q_1}\\ \mathrm{Q_2}\\ \vdots\\ \mathrm{Q_m}\end{matrix}\begin{bmatrix}\mathrm{v_{11}} & \mathrm{v_{12}} & \cdots & \mathrm{v_{17}}\\ \mathrm{v_{21}} & \mathrm{v_{22}} & \cdots & \mathrm{v_{27}}\\ \vdots & & \ddots & \vdots\\ \mathrm{v_{m1}} & \mathrm{v_{m2}} & \cdots & \mathrm{v_{m7}}\end{bmatrix}$$

v_{ik} is the membership degree of question Q_i at comprehensive level k.

Step4: Calculation the final adjustment value of each question

Based on matrix LM, calculate the weighted adjustment value AL_i of question Q_i. The formula for calculating is shown as follows:

$$AL_i = w_i(v_{i7} + v_{i6} + v_{i5} + v_{i4} - v_{i3} - v_{i2} - v_{i1})/7, \quad (14)$$

where $w_i = \frac{g_i}{\sum_i^m g_i}$ is the weight of a question grade. So far, we get a vector AL for the weighted levels of all questions,

Authorized licensed use limited to: Tejasvi Kolhe. Downloaded on July 12,2026 at 18:07:37 UTC from IEEE Xplore. Restrictions apply.

$$AL = \begin{matrix} Q_1 \\ Q_2 \\ \vdots \\ Q_m \end{matrix}\begin{bmatrix} al_1 \\ al_2 \\ \vdots \\ al_m \end{bmatrix}$$

Step5: Readjustment of students' grades

Using AL, we calculate a new accuracy rate matrix A' from A,

$$\mathrm{A}' = \begin{matrix} \mathrm{Q_1} \\ \mathrm{Q_2} \\ \vdots \\ \mathrm{Q_m} \end{matrix}\begin{bmatrix} \mathrm{a'_{11}} & \mathrm{a'_{12}} & \cdots & \mathrm{a'_{1n}} \\ \mathrm{a'_{21}} & \mathrm{a'_{22}} & \cdots & \mathrm{a'_{2n}} \\ \vdots & & \ddots & \vdots \\ \mathrm{a'_{m1}} & \mathrm{a'_{m2}} & \cdots & \mathrm{a'_{mn}} \end{bmatrix}$$

$$a'_{ij} = a_{ij} * (1 + al_i) \tag{15}$$

based on A' and G^T, we get a new average score for each student,

$$S' = G^T A' = \left[s'_j\right] \quad 1 \le j \le n. \tag{16}$$

So far, we can get more reasonable evaluation of students' performance.

IV. Exemplary application

In this section, a case has been proceeded to elaborate the process of student learning achievement evaluation by using the method proposed above. Given the accuracy rate matrix A and grade distribution matrix G are as follows,

$$\mathrm{A} = \begin{matrix} \mathrm{Q_1} \\ \mathrm{Q_2} \\ \mathrm{Q_3} \\ \mathrm{Q_4} \\ \mathrm{Q_5} \end{matrix}\begin{bmatrix} 0.59 & 0.35 & 1 & 0.66 & 0.11 & 0.08 & 0.84 & 0.23 & 0.4 & 0.24 \\ 0.01 & 0.27 & 0.14 & 0.04 & 0.88 & 0.16 & 0.04 & 0.22 & 0.81 & 0.53 \\ 0.77 & 0.69 & 0.97 & 0.71 & 0.17 & 0.86 & 0.87 & 0.42 & 0.91 & 0.74 \\ 0.73 & 0.72 & 0.18 & 0.16 & 0.5 & 0.02 & 0.32 & 0.92 & 0.9 & 0.25 \\ 0.93 & 0.49 & 0.08 & 0.81 & 0.65 & 0.93 & 0.39 & 0.51 & 0.97 & 0.61 \end{bmatrix}$$

$$\mathrm{G} = \begin{matrix} \mathrm{Q_1} \\ \mathrm{Q_2} \\ \mathrm{Q_3} \\ \mathrm{Q_4} \\ \mathrm{Q_5} \end{matrix}\begin{bmatrix} 10 \\ 15 \\ 20 \\ 25 \\ 30 \end{bmatrix}$$

The predetermined importance and complexity degree of each question are prepared in the following vectors PV and CV respectively,

$$\mathrm{PV} = \begin{matrix} \mathrm{Q_1} \\ \mathrm{Q_2} \\ \mathrm{Q_3} \\ \mathrm{Q_4} \\ \mathrm{Q_5} \end{matrix}\begin{bmatrix} 0.2 \\ 0.15 \\ 0.5 \\ 0.7 \\ 0.25 \end{bmatrix} \qquad \mathrm{CV} = \begin{matrix} \mathrm{Q_1} \\ \mathrm{Q_2} \\ \mathrm{Q_3} \\ \mathrm{Q_4} \\ \mathrm{Q_5} \end{matrix}\begin{bmatrix} 0.05 \\ 0.15 \\ 0.25 \\ 0.7 \\ 0.5 \end{bmatrix}$$

we can see that question Q_4 has highest importance and complexity among all questions. The difficulty vector DV is calculated using (4),

$$\mathrm{DV} = \begin{matrix} \mathrm{Q_1} \\ \mathrm{Q_2} \\ \mathrm{Q_3} \\ \mathrm{Q_4} \\ \mathrm{Q_5} \end{matrix}\begin{bmatrix} 0.55 \\ 0.69 \\ 0.29 \\ 0.53 \\ 0.36 \end{bmatrix}$$

Step 1: Based on (2), calculate the total results of each student,

$$\mathrm{S} = \begin{bmatrix} \mathrm{S_1} & \mathrm{S_2} & \mathrm{S_3} & \mathrm{S_4} & \mathrm{S_5} & \mathrm{S_6} & \mathrm{S_7} & \mathrm{S_8} & \mathrm{S_9} & \mathrm{S_{10}} \\ 67.6 & 54.05 & 38.4 & 49.7 & 49.7 & 48.8 & 46.1 & 52.3 & 85.95 & 49.7 \end{bmatrix}$$

we obtain the classical ranking of students' scores in a descending sequence:

$$\mathrm{S_9}>\mathrm{S_1}>\mathrm{S_2}>\mathrm{S_8}>\mathrm{S_4}=\mathrm{S_5}=\mathrm{S_{10}}>\mathrm{S_6}>\mathrm{S_7}>\mathrm{S_3}.$$

It is clear that student S_4, S_5 and S_{10} have the same score of 49.7.

Step 2: Fuzzify the importance vector PV, complexity vector CV, difficulty vector DV, we calculate the fuzzy importance matrix PM, complexity matrix CM, and difficulty matrix DM, respectively.

$$\mathrm{PM} = \begin{matrix} \mathrm{Q_1} \\ \mathrm{Q_2} \\ \mathrm{Q_3} \\ \mathrm{Q_4} \\ \mathrm{Q_5} \end{matrix}\begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0.33 & 0.5 \\ 0.75 & 0.17 & 0 \end{bmatrix}$$

$$\mathrm{CM} = \begin{matrix} \mathrm{Q_1} \\ \mathrm{Q_2} \\ \mathrm{Q_3} \\ \mathrm{Q_4} \\ \mathrm{Q_5} \end{matrix}\begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \\ 0.75 & 0.17 & 0 \\ 0 & 0.33 & 0.5 \\ 0 & 1 & 0 \end{bmatrix}$$

$$\mathrm{DM} = \begin{matrix} \mathrm{Q_1} \\ \mathrm{Q_2} \\ \mathrm{Q_3} \\ \mathrm{Q_4} \\ \mathrm{Q_5} \end{matrix}\begin{bmatrix} 0 & 0.83 & 0 \\ 0 & 0.37 & 0.45 \\ 0.56 & 0.3 & 0 \\ 0 & 0.9 & 0 \\ 0.18 & 0.54 & 0 \end{bmatrix}$$

Step 3: Through the above specific matrix PM, CM and DM, fuzzy reasoning can be executed with the fuzzy rules from R1 to R25 and (7)-(13), the comprehensive level matrix LM is finally obtained,

$$\mathrm{LM} = \begin{matrix} \mathrm{Q_1} \\ \mathrm{Q_2} \\ \mathrm{Q_3} \\ \mathrm{Q_4} \\ \mathrm{Q_5} \end{matrix}\begin{bmatrix} 0.7 & 0.95 & 0.7 & 0.65 & 0.4 & 0.25 & 0 \\ 0.7 & 0.81 & 0.84 & 0.54 & 0.54 & 0.14 & 0.14 \\ 0.39 & 0.79 & 0.71 & 0.62 & 0.49 & 0.4 & 0 \\ 0 & 0.27 & 0.4 & 0.5 & 0.57 & 0.62 & 0.35 \\ 0.36 & 0.66 & 0.76 & 0.6 & 0.46 & 0.3 & 0 \end{bmatrix}$$

Step 4: Combined matrix LM with (14), calculate the weighted adjustment level of question Q_i,

$$AL = \begin{matrix} Q_1 \\ Q_2 \\ Q_3 \\ Q_4 \\ Q_5 \end{matrix}\begin{bmatrix} -0.02 \\ -0.02 \\ -0.01 \\ +0.05 \\ -0.02 \end{bmatrix} \tag{17}$$

Step 5: Based on AL and (15), we calculate the new accuracy rate matrix A'.

$$\mathrm{A}' = \begin{matrix} \mathrm{Q_1} \\ \mathrm{Q_2} \\ \mathrm{Q_3} \\ \mathrm{Q_4} \\ \mathrm{Q_5} \end{matrix}\begin{bmatrix} 0.58 & 0.34 & 0.98 & 0.65 & 0.11 & 0.08 & 0.83 & 0.23 & 0.39 & 0.24 \\ 0.01 & 0.26 & 0.14 & 0.04 & 0.86 & 0.16 & 0.04 & 0.22 & 0.79 & 0.52 \\ 0.76 & 0.68 & 0.96 & 0.70 & 0.17 & 0.85 & 0.86 & 0.42 & 0.90 & 0.73 \\ 0.77 & 0.75 & 0.19 & 0.17 & 0.52 & 0.02 & 0.34 & 0.96 & 0.94 & 0.26 \\ 0.91 & 0.48 & 0.08 & 0.79 & 0.64 & 0.91 & 0.38 & 0.50 & 0.95 & 0.60 \end{bmatrix}$$

Authorized licensed use limited to: Tejasvi Kolhe. Downloaded on July 12,2026 at 18:07:37 UTC from IEEE Xplore. Restrictions apply.

Based on (2), we recalculate each student's total score,

$$S = \begin{bmatrix} S_1 & S_2 & S_3 & S_4 & S_5 & S_6 & S_7 & S_8 & S_9 & S_{10} \\ 67.7 & 54.4 & 38.2 & 49.2 & 49.6 & 48.1 & 46 & 53 & 86 & 49.3 \end{bmatrix}$$

we get a new ranking order for S_4, S_5 and S_{10}: $S_5 > S_{10} > S_4$. The final ranking order for all students becomes:

$$S_9 > S_1 > S_2 > S_8 > S_5 > S_{10} > S_4 > S_6 > S_7 > S_3$$

the modified grades did not change the original ranking order, but provided distinct ranking for students with equal final scores.

In order to quantitatively analyze the effects of question's attitudes on the evaluation results, we calculate the percentage difference between adjusted accuracy rates and original accuracy rates, based on (15), the formula is shown as follows,

$$\alpha_{ij} = \frac{a'_{ij} - a_{ij}}{a_{ij}} = al_i \tag{18}$$

where $1 \le i \le 5$, $1 \le j \le 10$ for the index of questions and students, respectively. So the vector *AL* in (17) reflects the relative difference. We can see that the adjusted accuracy rates are increased only for Q_4, and are decreased for other questions. Through comparison of the data in *PV*, *CV*, and *DV*, we find that the pre-set importance and complexity of Q_4 are the biggest among all questions, which naturally results in the weighted level of Q_4 is the highest, +0.05. From this analysis, we can detect the effects of question's attributes: importance, complexity and difficulty on the ranking order of students, because if one student obtains high accuracy rate in Q_4, which means he masters the corresponding knowledge better than others. Since this piece of knowledge has higher importance and complexity, and may be difficulty, it is reasonable for this student to obtain a higher ranking position. It is the goal that we hope to achieve for fair and transparent evaluation of students' learning performance.

V. Conclusions

In traditional evaluation system, exam scores are used to classify the students' learning performance. In this paper, by considering three aspect of a question, i.e. importance, complexity and difficulty, a fuzzy based inference approach is used to establish seven levels for each question. Using these levels of all questions, new accuracy rates of students are calculated from original accuracy rates. Based on the new accuracy rates, the students who have equal total scores can be completely distinguished, and without influencing the ranking order of other students. The proposed method in this paper provide more transparent and fair assessment and enriches the present system of students' performance assessment, and is utilizable to analyze the direct relationship of students' scores with the attributes of each questions, more clearly reflect the teaching design and intention of each question. The results produced by the method are easy communicate with teaching staff, and can be regularly inspected and strengthened.

Acknowledgment

This work is supported by "Guangdong Innovation Program for Postgraduate Education (No.2017SFKC-01)".

References

[1] Zadeh, L. A.. Fuzzy sets. Information and Control, 1965, 8, 338–353.

[2] S. Weon and J. Kim, "Learning achieveemnt evaluation strategy using fuzzy membership function," In proceedings of the 31st ASEE/IEEE fronties in education conference,2001, Reno, 1, 19-24.

[3] S. M. Bai and S. M. Chen, "Evaluating students' learning achievement using fuzzy membership functions and fuzzy rules," Expert Systems with Applications, 2008, 34(1):399-410.

[4] S. M. Bai and S. M. Chen, "Automatically constructing grade membership functions of fuzzy rules for students' evaluation," Expert Systems with Applications, 2008, 35(3):1408-1414.

[5] Biswas R. , "An application of fuzzy sets in students' evaluation," Fuzzy Sets and Systems, 1995, 74(2):187-194.

[6] K. C. Chai, K. M. Tay, and C. P. Lim , "A new fuzzy peer assessment methodology for cooperative learning of students," Applied Soft Computing, 2015, 32:468-480.

[7] D. F. Chang and C. M. Sun, "Fuzzy assessment of learning performance of junior high school students," In Proceedings of the 1993 first national symposium on fuzzy theory and applications,1993, 1-10.

[8] S. M. Chen and C. H. Lee, "New methods for students' evaluating using fuzzy sets," Fuzzy Sets and Systems, 104(2), pp.209–218.

[9] S. M. Chen and T. K. Li, " Evaluating students' learning achievement based on fuzzy rules with fuzzy reasoning capability," Expert Systems with Applications, 2011, 38(4):4368-4381.

[10] S. J. Chen and C. L. Hwang, Fuzzy Multiple Attribute Decision Making[M]. Springer Berlin Heidelberg, 1992.

[11] G. Gokmen, T. C. Akinci, M. Tekta, N. Onat, G. Kocyigit, and N. Tekta, "Evaluation of Student Performance in Laboratory Applications Using Fuzzy Logic," Procedia-Social and Behavioral Sciences, 2010,2(2): 902–909.

[12] I. A. Hameed, "A Fuzzy System to Automatically Evaluate and Improve Fairness of Multiple-Choice Questions (MCQs) based Exams," In Proceedings of the 8th International Conference on Computer Supported Education (CSEDU 2016), 2016,pp.476-481.

[13] I. A. Hameed, " Using Gaussian membership functions for improving the reliability and robustness of students' evaluation systems," Expert Systems with Applications, 2011, 38(6):7135-7142.

[14] K. Chrysafiadi and M. Virvou, "Evaluating the integration of fuzzy logic into the student model of a web-based learning environment," Expert Systems with Applications, 2012, 39(18):13127–13134.

[15] C. K. Law, "Using fuzzy numbers in educational grading system," Fuzzy Sets and Systems,1996, 83(3):311-323.

[16] J. Ma and D. Zhou, "Fuzzy set approach to the assessment of student-centered learning," IEEE Transactions on Education,2000, 43(2):237-241.

[17] M. Samarakou, A. Papadakis, P. Prentakis, D. Karolidis & S. Athineos, " A fuzzy model for enhanced student evaluation," The International Journal of Learning Annual Review, 2009, 16(10), 103-118.

[18] R. S. Yadav, V. P. Singh, "Modeling Academic Performance Evaluation Using Soft Computing Techniques: A Fuzzy Logic Approach," International Journal on Computer Science and Engineering, 2011, 3(2):676-686.

[19] I. Saleh and S. I. Kim, "A fuzzy system for evaluating students' learning achievement.," Expert Systems with Applications ,2009,36(3):6236-6243.

[20] H. Y. Wang and S. M. Chen, "Evaluating Students' Answerscripts Using Fuzzy Numbers Associated With Degrees of Confidence," IEEE Transactions on Fuzzy Systems, 2008, 16(2):403-415.

[21] S. Weon and J. Kim, "Learning achievement evaluation strategy using fuzzy membership function," In proceedings of the 31st ASEE/IEEE fronties in education conference, 2001,Reno, 1, 19-24.

[22] E. Wilson, C. L. Karr and L. M. Freeman, "Flexible, adaptive, automatic fuzzybased grade assigning system," In Proceedings of the North American fuzzy information processing society conference, 1998, pp.334–338.

Authorized licensed use limited to: Tejasvi Kolhe. Downloaded on July 12,2026 at 18:07:37 UTC from IEEE Xplore. Restrictions apply.